

Experiment No.	1
Experiment Title	Exploratory Data Analysis using Machine Learning Algorithms
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Submission Date	26th July 2026

1 Objective

The aim of this experiment is to develop a reusable and general-purpose machine learning pipeline that can be applied to any tabular dataset. The pipeline automates the key stages of a machine learning workflow, including data preprocessing, exploratory data analysis, feature selection, model training, and performance evaluation. This experiment helps in understanding how each stage contributes to building accurate and reliable machine learning models while reducing the need to write dataset-specific code.

2 Problem Statement

The objective of this problem is to predict whether a person has diabetes based on common health and demographic information such as age, gender, BMI, blood glucose level, HbA1c level, hypertension, and heart disease. This is a binary classification task where the target variable diabetes is 1 for diabetic patients and 0 otherwise. The experiment also compares multiple classification algorithms on the same dataset to determine which model provides the best prediction performance.

3 Dataset Description

Dataset Name	Diabetes Prediction Dataset
Dataset Source	Kaggle- <code>diabetes_prediction_dataset.csv</code>
Number of Samples	100,000 (raw); 96,146 after removing duplicate rows
Number of Features	8 input features (gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level) plus 1 target column (diabetes)
Number of Classes	2 (diabetic = 1, non-diabetic = 0)
Missing Values	None found
Train-Test Split	70% train / 10% validation / 20% test, stratified on the target

The dataset is reasonably large (close to 100,000 records to start with) and mixes numerical clinical measurements (age, BMI, HbA1c level, blood glucose level) with categorical demographic and lifestyle fields (gender, smoking history) and two binary medical flags (hypertension, heart disease). One thing worth noting immediately is that the

classes are imbalanced: only around 8.8% of patients in the cleaned dataset are labelled as diabetic, so accuracy alone can be a little misleading and needs to be read alongside precision, recall and F1-score.

4 Exploratory Data Analysis

Before any modelling was attempted, the dataset was passed through a generic `perform_eda()` routine that prints summary statistics and produces a standard battery of plots. Since the pipeline label-encodes the two categorical columns (`gender` and `smoking_history`) and then standardises every numeric column during preprocessing, all nine columns appear as continuous, zero-mean, unit-variance values by the time the EDA plots are generated this is why the axes in the plots below are centred around zero rather than showing the original raw values.

Missing Values Heatmap

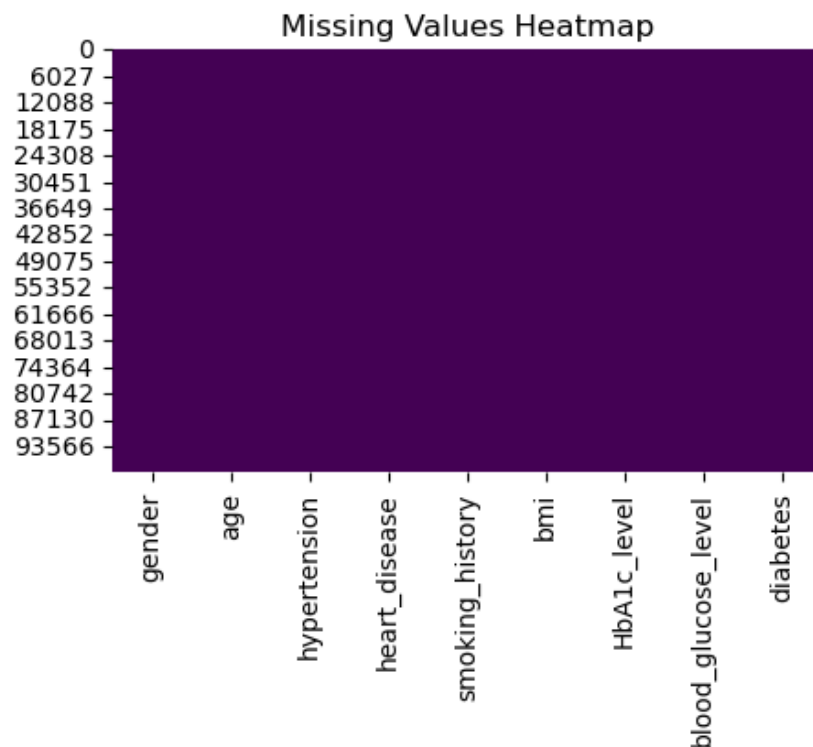


Figure 1: Missing values heatmap for the diabetes dataset.

Inference: The heatmap shows that there are no missing values in the dataset. This means the data is complete and we need not handle the missing values.

Correlation Heatmap

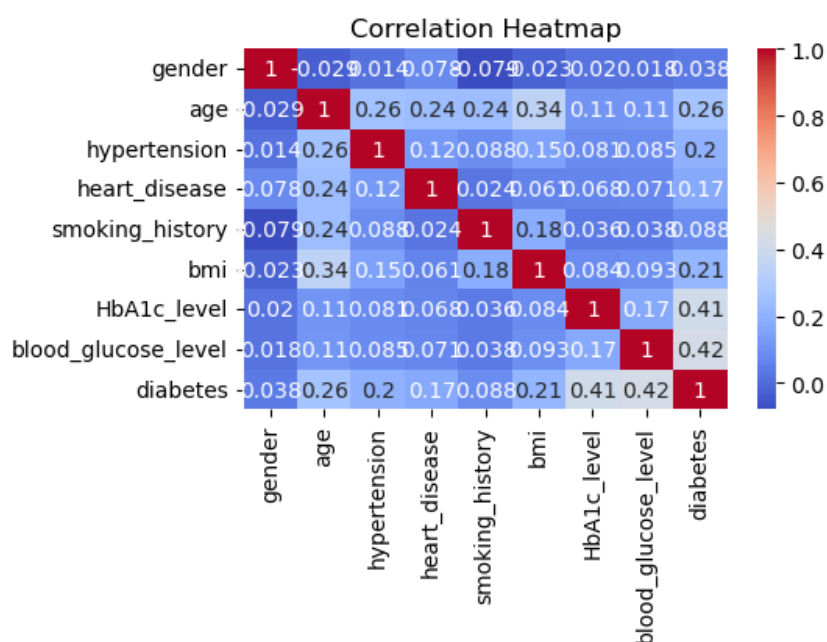


Figure 2: Correlation heatmap of the numeric (encoded) features.

Inference: Blood glucose level and HbA1c level have the strongest relationship with diabetes. Age and BMI also show some correlation, while the remaining features have a weaker effect.

Histograms of Numeric Features

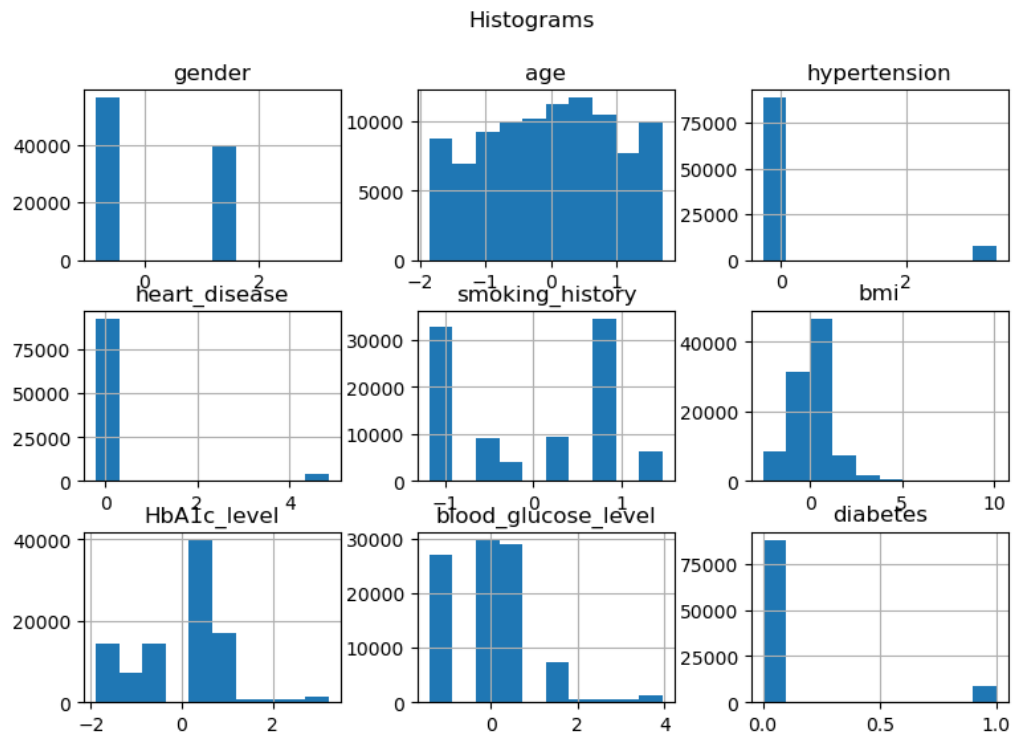


Figure 3: Histograms of all nine numeric (post-encoding, post-scaling) columns.

Inference: Most patients in the dataset do not have diabetes, showing that the data is imbalanced. Age and BMI are spread across different values, while hypertension and heart disease mostly have one common value.

Boxplots

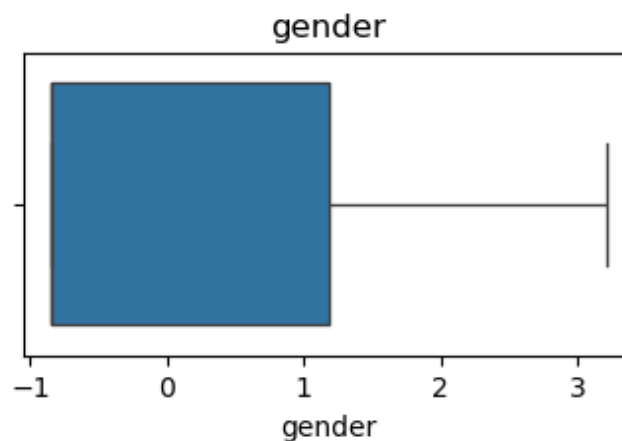


Figure 4: Boxplot - gender.

Inference: The gender column contains only the expected categories. Since it is a categorical feature, the boxplot does not provide much information other than confirming

the data is valid.

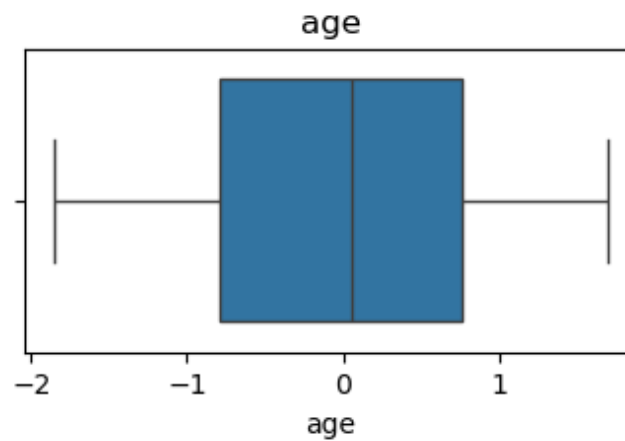


Figure 5: Boxplot - age.

Inference: The age values are spread across a wide range with no major outliers. This shows that the dataset includes patients from different age groups

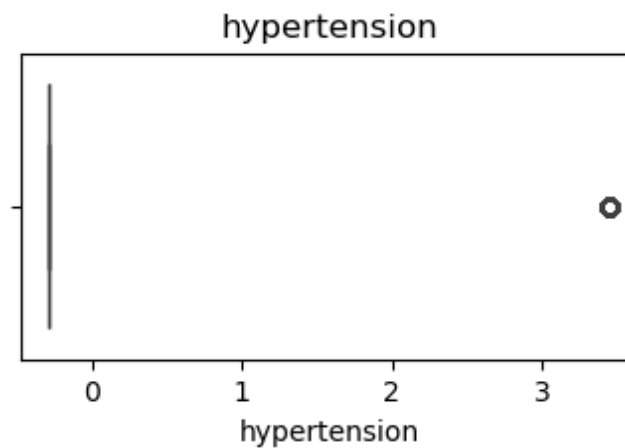


Figure 6: Boxplot - hypertension.

Inference: Most patients do not have hypertension, while only a small number do. The values are valid and no incorrect entries are present.

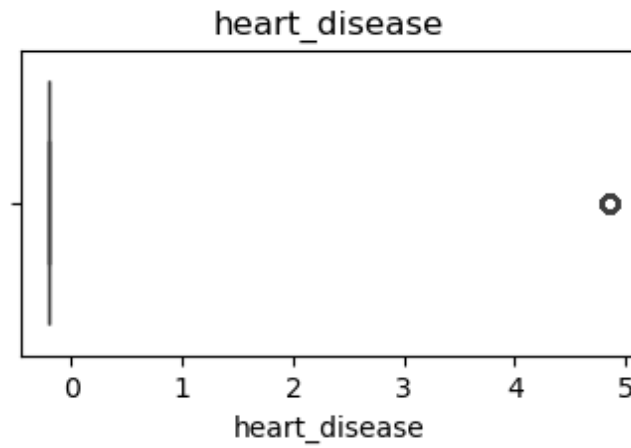


Figure 7: Boxplot - heart_disease.

Inference: Most patients do not have heart disease. The column contains only the expected values and does not require any additional cleaning.

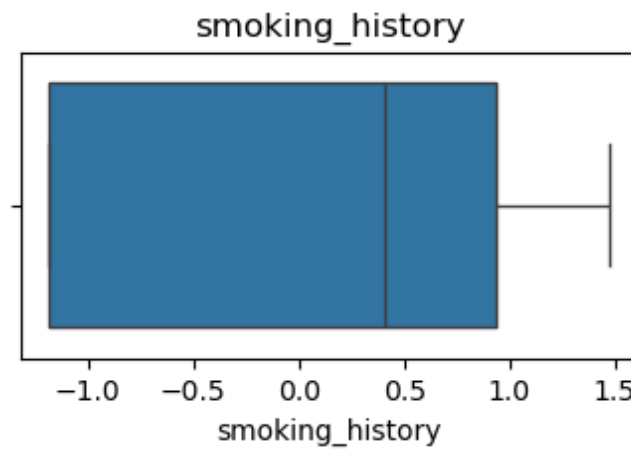


Figure 8: Boxplot - smoking_history.

Inference: Patients have different smoking histories, so the values are spread across several categories. This provides useful information for prediction.

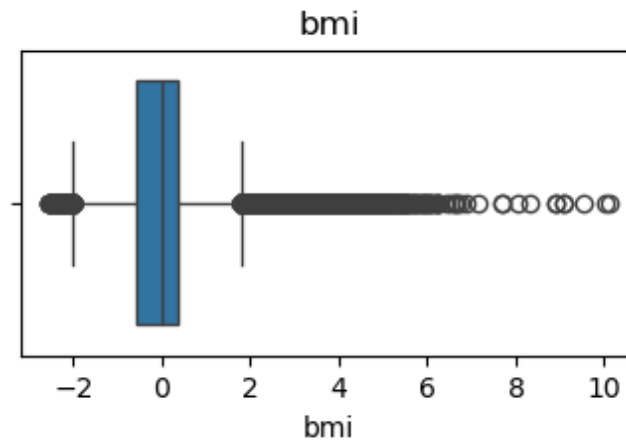


Figure 9: Boxplot - bmi.

Inference: BMI contains a few high values, but they are likely to be genuine patient records. These values were kept because they may help the model identify diabetes more accurately.

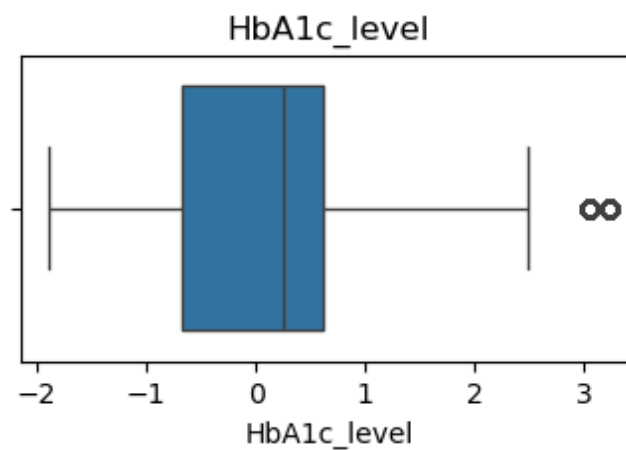


Figure 10: Boxplot - HbA1c.level.

Inference: Most HbA1c values fall within a normal range, with a few higher values. These higher values are important because they are closely related to diabetes.

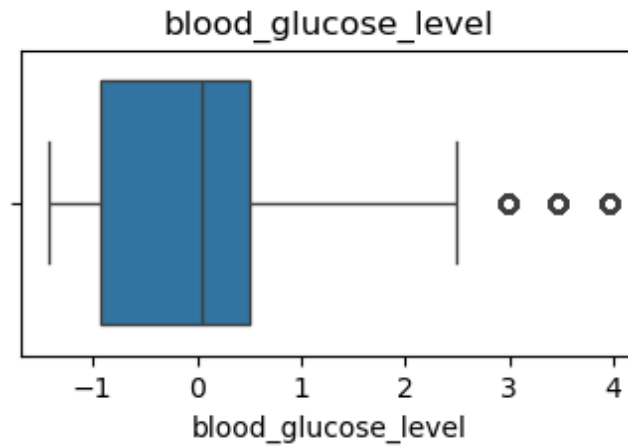


Figure 11: Boxplot - blood_glucose_level.

Inference: A few patients have high blood glucose levels, which are likely to belong to diabetic patients. These values were kept as they are useful for prediction.

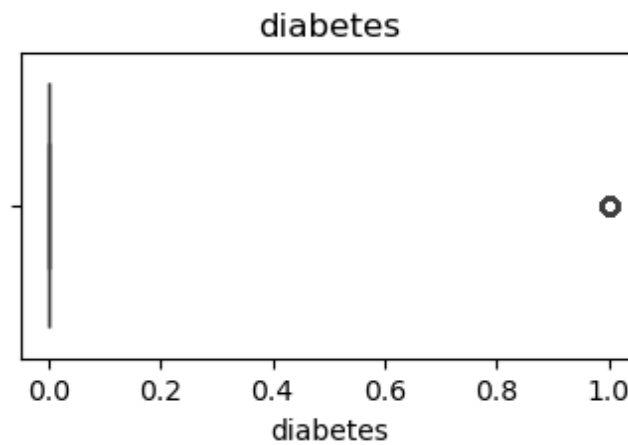


Figure 12: Boxplot - diabetes (target variable).

Inference: The plot shows that there are more non-diabetic patients than diabetic patients. This class imbalance should be considered when evaluating the models.

Scatter Plot

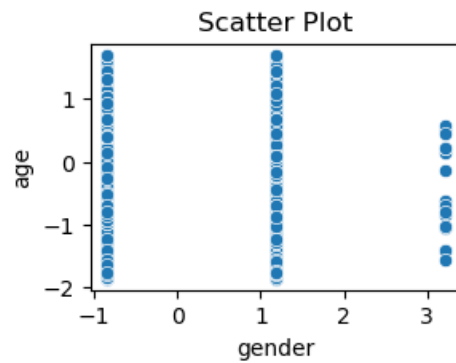


Figure 13: Scatter plot of the first two numeric columns, gender vs. age.

Inference: The scatter plot shows the relationship between gender and age. It confirms that patients of different ages are present in each gender category.

Density (KDE) Plots

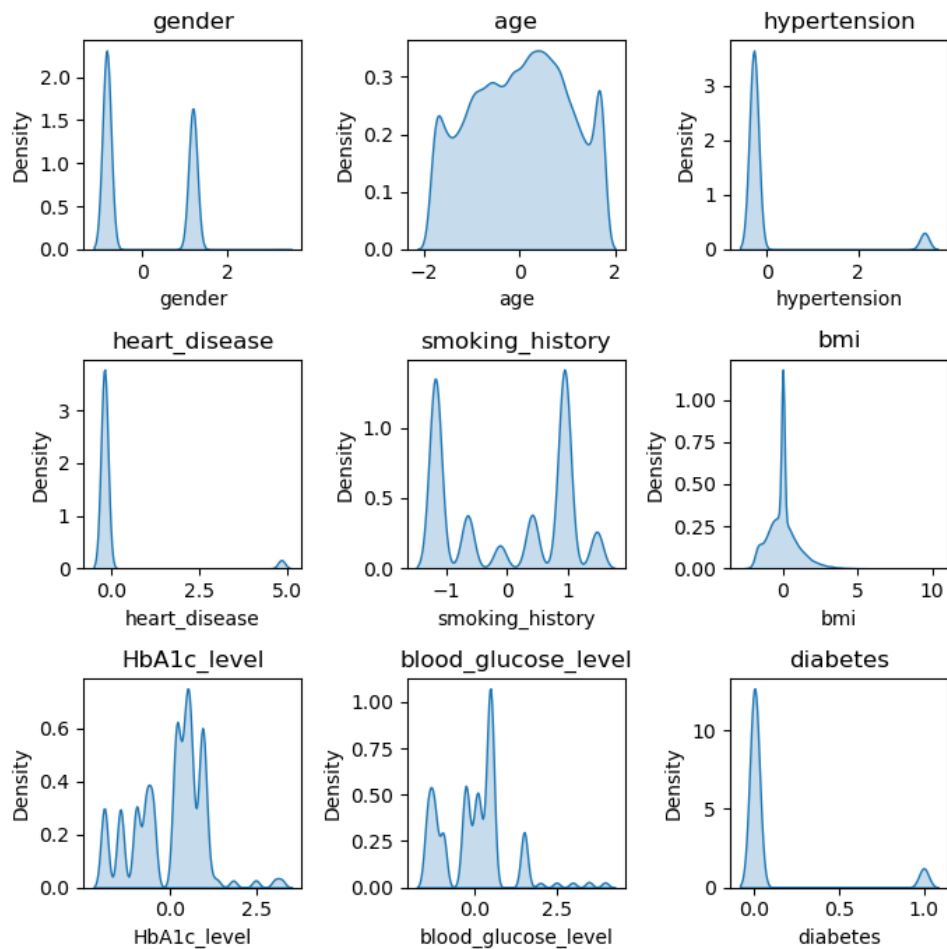


Figure 14: Kernel density estimate plots for all nine numeric columns.

Inference: The density plots show how the values are distributed for each feature. Blood glucose level and HbA1c level show clear patterns, indicating that they are important features for predicting diabetes.

5 Data Preprocessing

All preprocessing was handled by a single reusable function, `load_and_preprocess()`, so that the same steps could later be applied to other datasets without rewriting the logic. The steps carried out, in order, were:

- **Remove duplicate**
- **Empty row/column removal**
- **Missing value handling**
- **Whitespace and type cleanup**
- **Constant column removal**
- **Categorical encoding**
- **Feature scaling:** All numeric feature columns were standardised, so that every feature has zero mean and unit variance this is what gives the EDA plots their centred, symmetric appearance around zero.
- **Train/validation/test split**

6 Machine Learning Algorithm

Logistic Regression

Logistic Regression models the probability of the positive class using the sigmoid function applied to a linear combination of the input features, $P(y = 1 | x) = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}$, with the weights learned by maximising the log-likelihood of the observed labels. It is simple, fast to train, and gives directly interpretable coefficients, but it can only capture linear decision boundaries and tends to underperform when the true relationship between features and target is more complex or non-linear.

Decision Tree

A Decision Tree recursively splits the feature space into regions by choosing, at each node, the feature and threshold that most reduces impurity (commonly measured with the Gini index, $Gini = 1 - \sum_i p_i^2$). It is easy to interpret and requires little data preparation, but a single tree grown without constraints tends to overfit the training data and can be unstable – small changes in the data can produce a very different tree.

Random Forest

Random Forest is an ensemble of many decision trees, each trained on a bootstrapped sample of the data and a random subset of features, with the final prediction taken as a majority vote across the trees. Averaging over many de-correlated trees substantially reduces the overfitting problem of a single tree and generally improves accuracy, at the cost of reduced interpretability and higher computational and memory cost compared to a single tree.

K-Nearest Neighbours (KNN)

KNN is a non-parametric method that classifies a new point by looking at the majority class among its k nearest neighbours in the (scaled) feature space, typically using Euclidean distance. It makes no assumptions about the underlying data distribution and can capture complex decision boundaries, but it is sensitive to the choice of k , sensitive to feature scaling (which is why standardisation was applied beforehand), and can become slow to query as the dataset grows large.

Support Vector Machine (SVM)

An SVM finds the hyperplane that maximises the margin between the two classes, $\min \frac{1}{2} \|w\|^2$ subject to $y_i(w^\top x_i + b) \geq 1$, and can use kernel functions to handle non-linearly separable data. It tends to perform well on high-dimensional data and is relatively robust to overfitting when properly regularised, but it can be computationally expensive to train on large datasets and its performance is sensitive to the choice of kernel and regularisation parameters.

Naive Bayes

Gaussian Naive Bayes applies Bayes' theorem, $P(y | x) \propto P(y) \prod_i P(x_i | y)$, under the (naive) assumption that features are conditionally independent given the class, modelling each feature's class-conditional distribution as Gaussian. It is extremely fast to train and works reasonably well even with relatively little data, but the independence assumption rarely holds exactly in practice, which can hurt performance on datasets like this one where features such as HbA1c and blood glucose are clinically related to one another.

7 Experimental Setup

Python Version	3.12
Scikit-Learn Version	1.6.1
IDE	Jupyter Notebook
Operating System	Linux
Hardware	College Lab

8 Hyperparameter Selection

The aim of this experiment was to compare different machine learning algorithms using the same settings. Therefore, most models were trained using the default hyperparame-

ters provided by scikit-learn. The only change made was increasing the `max_iter` value of Logistic Regression to 1000 so that the model could converge properly. For SVM, `probability=True` was enabled to obtain probability predictions. Feature selection was performed using the ANOVA F-test with `SelectKBest`, and all eight available features were selected.

Parameter	Value
Logistic Regression – <code>max_iter</code>	1000
Decision Tree	Default settings
Random Forest – <code>n_estimators</code>	100
KNN – <code>n_neighbors</code>	5
SVM	RBF kernel, <code>probability=True</code>
Naive Bayes	Default settings
Feature Selection	ANOVA F-test, $k = 8$

A natural extension of this experiment would be to run a grid or randomised search over each model’s key hyperparameters (for example `n_estimators` and `max_depth` for Random Forest, or `C` and `gamma` for the SVM) – this was intentionally left out here so the comparison reflects each algorithm’s “out of the box” behaviour on this dataset.

9 Results

Before training, the ANOVA F-test feature selection step ranked all eight input features by how well they separate the two classes:

Feature	F-Score
blood_glucose_level	21113.49
HbA1c_level	19021.65
age	7257.36
bmi	4656.55
hypertension	3829.20
heart_disease	2885.95
smoking_history	758.46
gender	136.21

Since only eight features were available and $k = 8$ was requested, all eight were retained; the ranking is still useful, however, because it tells us which features are actually carrying the predictive weight. On the held-out test set of 19,230 patients, the six classifiers performed as follows (sorted by accuracy):

Metric	Value (Random Forest – best model)
Accuracy	0.9696
Precision (weighted)	0.9691
Recall (weighted)	0.9696
F1-score (weighted)	0.9675
ROC-AUC	Not computed in this run

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.9696	0.9691	0.9696	0.9675
SVM	0.9631	0.9632	0.9631	0.9591
KNN	0.9605	0.9585	0.9605	0.9576
Logistic Regression	0.9589	0.9564	0.9589	0.9561
Decision Tree	0.9494	0.9509	0.9494	0.9501
Naive Bayes	0.9028	0.9211	0.9028	0.9102

Random Forest achieved the highest accuracy and performed the best among all the models. SVM and KNN also gave very good results, while Logistic Regression and Decision Tree performed slightly lower. Naive Bayes had the lowest accuracy because its assumptions are less suitable for this dataset. Overall, Random Forest was the most effective model for this prediction task.

10 Result Visualization

Feature Selection Scores

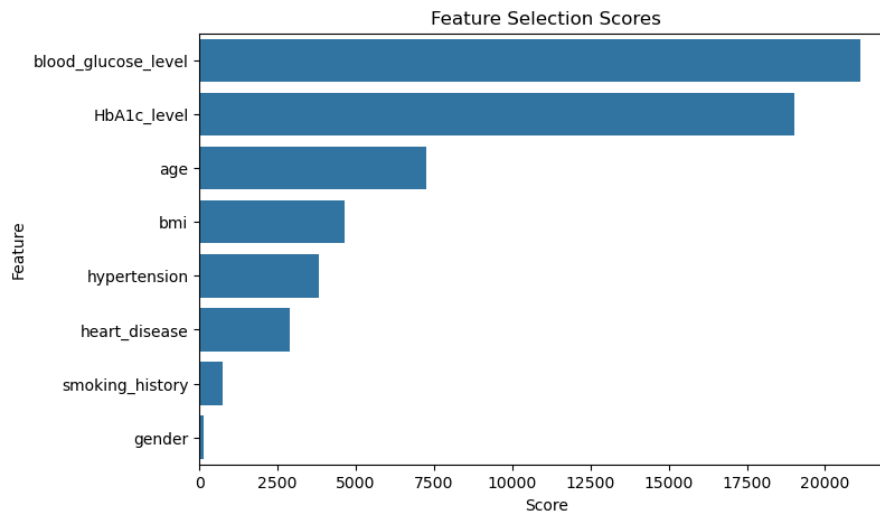


Figure 15: ANOVA F-scores for each feature, used to rank predictive importance.

Inference: Blood glucose level and HbA1c level have the highest feature scores, showing that they are the most important features for predicting diabetes. Age and BMI also contribute well, while gender has the lowest impact on the prediction.

Confusion Matrices

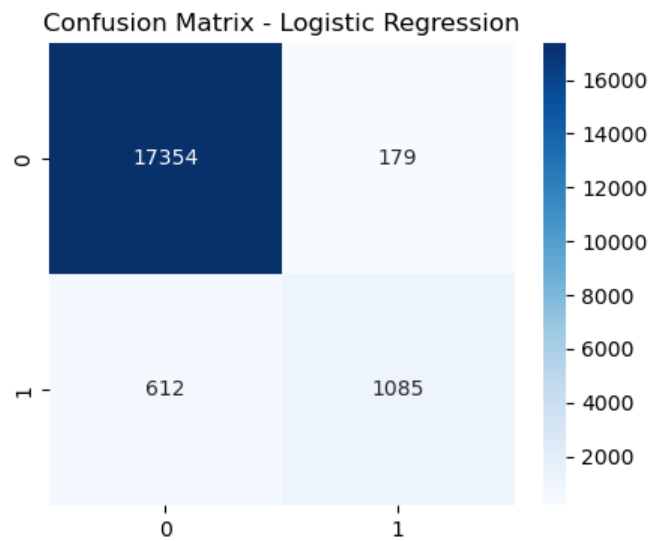


Figure 16: Confusion Matrix – Logistic Regression.

Inference: Logistic Regression correctly predicts most non-diabetic patients but misses some diabetic cases. Overall, it gives good results but is less effective than the best performing models.

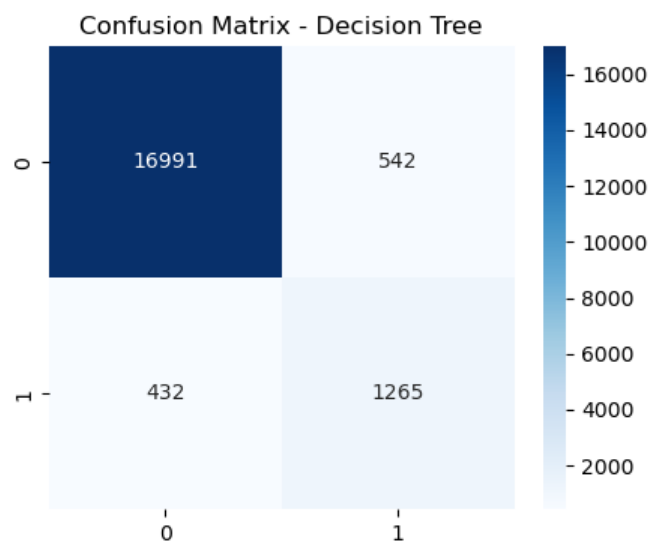


Figure 17: Confusion Matrix - Decision Tree.

Inference: The Decision Tree performs reasonably well but makes more prediction errors than Random Forest. This shows that a single decision tree is less accurate than group of trees.

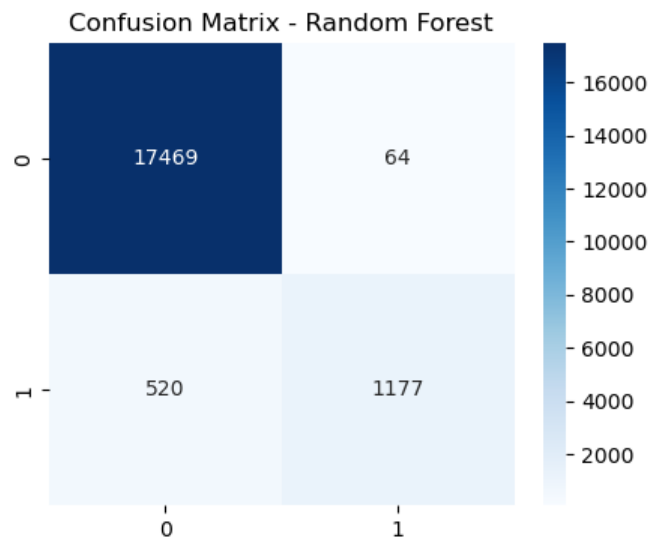


Figure 18: Confusion Matrix - Random Forest.

Inference: Random Forest has the fewest prediction errors among all the models. It correctly classifies most patients, making it the best-performing model in this experiment.

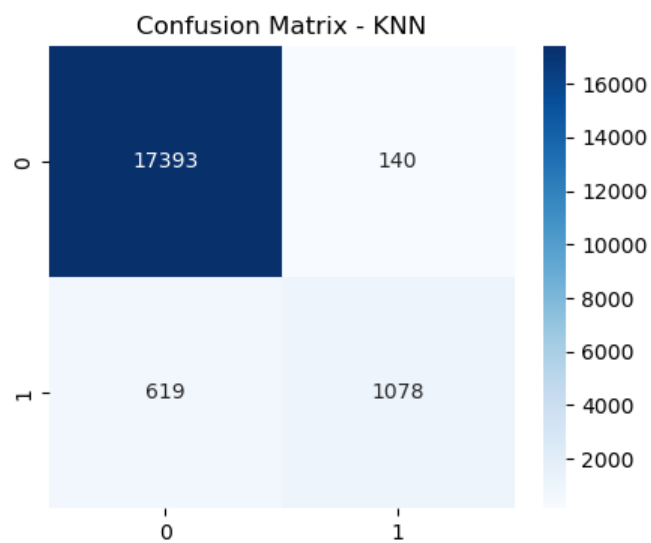


Figure 19: Confusion Matrix – KNN.

Inference: KNN performs well and gives results close to Random Forest and SVM. It makes only a few incorrect predictions and provides good overall accuracy.

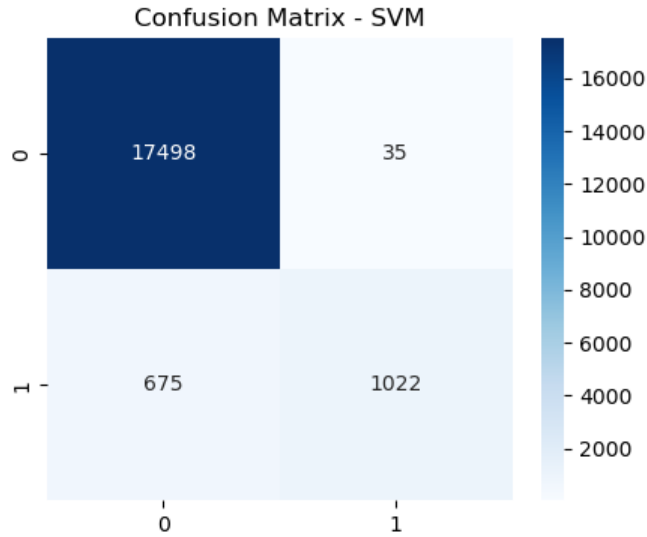


Figure 20: Confusion Matrix – SVM.

Inference: SVM also performs very well with only a small number of incorrect predictions. It is one of the best models, although slightly behind Random Forest.

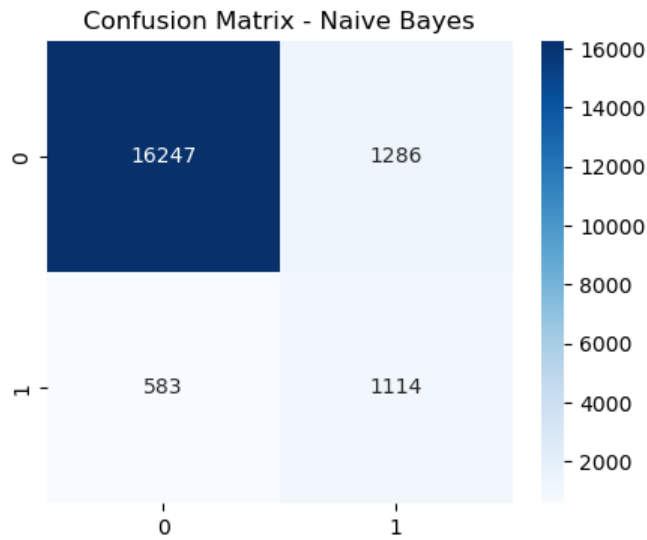


Figure 21: Confusion Matrix – Naive Bayes.

Inference: Naive Bayes has the highest number of incorrect predictions among all the models. It performs reasonably well but is less accurate than the other classifiers.

11 Discussion

The results show that blood glucose level and HbA1c level are the most important features for predicting diabetes. Among all the models, Random Forest achieved the best performance, followed by SVM and KNN. Logistic Regression also performed well, while Naive Bayes gave the lowest accuracy.

One limitation of this experiment is that the dataset is imbalanced, with fewer diabetic patients than non-diabetic patients. Also, most models were trained using their default settings, so their performance could be improved by tuning the hyperparameters.

In the future, the models can be improved by balancing the dataset, tuning the hyperparameters, calculating additional evaluation metrics such as ROC-AUC, and using one-hot encoding for categorical features.

12 Conclusion

This experiment successfully developed a complete machine learning pipeline for predicting diabetes. Blood glucose level and HbA1c level were found to be the most important features. Among the six models tested, Random Forest achieved the highest accuracy, followed by SVM and KNN. The experiment also showed that good data preprocessing, feature selection, and comparing multiple models are important steps in building an accurate and reliable machine learning model.

13 References

- [1] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [2] W. McKinney, “Data Structures for Statistical Computing in Python,” in *Proc. 9th Python in Science Conf.*, 2010, pp. 56–61.
- [3] Kaggle, “Diabetes Prediction Dataset,” [Online]. Available: <https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>.